

Neeraj Kumar Gontala

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SUMMARY

AI Engineer with 3+ years of experience building and deploying production-grade Machine Learning, Generative AI, and NLP solutions across financial services and enterprise applications. Experienced in designing Retrieval-Augmented Generation (RAG) systems, LLM-powered assistants, semantic search platforms, recommendation engines, and intelligent automation using Python, PyTorch, Hugging Face, LangChain, and FastAPI. Skilled in developing scalable ML pipelines, model deployment, MLOps, vector search, and cloud-native AI solutions on AWS and Azure. Proven ability to improve model accuracy, automate business workflows, and deliver secure, production-ready AI applications.

TECHNICAL SKILLS

Programming Languages:	Python (Advanced), SQL (PostgreSQL, BigQuery, Snowflake, Redshift), Flask, Streamlit, Bash
AI & Machine Learning:	Machine Learning, Deep Learning, NLP, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Optimization, Time Series Forecasting, Recommendation Systems, Anomaly Detection, Explainable AI
Generative AI & LLMs:	Large Language Models (LLMs), LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, LoRA, QLoRA, PEFT, BERT, RoBERTa, GPT Models, Semantic Search, Embeddings
Frameworks & Libraries:	PyTorch, TensorFlow, Scikit-learn, FastAPI, Flask, Django
Data Engineering:	PySpark, Apache Spark, Apache Airflow, Kafka, Databricks, ETL Pipelines
MLOps & Deployment:	MLflow, Docker, Kubernetes, Jenkins, CI/CD, Git, Model Monitoring, Drift Detection
Cloud Platforms:	AWS (S3, EC2, Lambda, SageMaker), Azure ML, Azure Blob Storage, GCP BigQuery
Databases:	PostgreSQL, MySQL, MongoDB
Visualization:	Power BI, Tableau

PROFESSIONAL EXPERIENCE

McKinsey and Company, USA

AI Engineer

Feb 2026 – Present

- Designed and deployed an enterprise Financial Intelligence Assistant using Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, and vector search, improving response accuracy to 94% while reducing customer query resolution time by 65%.
- Built production-grade conversational AI applications for customer support, financial document search, and intelligent knowledge retrieval using BERT, RoBERTa, and transformer-based NLP models, supporting 50,000+ monthly users.
- Developed semantic search pipelines using transformer embeddings, vector indexing, and retrieval optimization to provide sub-second access to financial policies, regulatory documents, contracts, and enterprise knowledge repositories.
- Engineered agentic AI workflows integrating tool calling, function orchestration, retrieval pipelines, and enterprise APIs to automate financial research and business decision support.
- Collaborated with software engineers, product teams, and business stakeholders to deploy AI capabilities into enterprise applications, establish model evaluation frameworks, and improve reliability, scalability, and user experience.

Tech Mahindra, India

Junior AI Engineer

Jan 2021 – Jul 2023

- Developed machine learning models using XGBoost, LightGBM, and Scikit-learn for pricing optimization, customer segmentation, and predictive analytics, improving pricing accuracy and increasing revenue by 18%.
- Designed and deployed end-to-end AI pipelines on AWS and Azure using Docker, MLflow, Jenkins, and FastAPI, reducing deployment cycles by 35% while enabling automated model retraining and monitoring.
- Built recommendation engines using customer transaction history, behavioral analytics, and financial profiles to deliver personalized product recommendations, increasing digital product adoption and customer engagement.
- Developed Retrieval-Augmented Generation (RAG) applications using LangChain, Hugging Face Transformers, embeddings, and vector search to automate document retrieval, knowledge discovery, and enterprise workflow automation.
- Implemented semantic search pipelines using embedding models and retrieval optimization techniques, improving document relevance and reducing irrelevant search results by 22%.
- Built deep learning models using PyTorch and TensorFlow for computer vision, NLP, and speech processing applications, achieving prediction accuracy above 92% across multiple production use cases.
- Established MLOps workflows with MLflow for experiment tracking, model versioning, explainability, lineage, and drift monitoring to support responsible AI and regulatory compliance.
- Optimized enterprise AI applications through prompt engineering, inference optimization, and model performance tuning, reducing infrastructure costs while improving response quality and application scalability.

EDUCATION

M.S. in Computer Science | Auburn University at Montgomery, AL, USA

Dec 2025